

Mean Field Continued

Global setup and clause distribution

Let K be the number of binary variables and M be the total number of clauses. We define the clause density parameter as $\alpha = M/K$.

Half of the clauses are AND and other half are XOR clauses. Note this fraction can be changed.

Let $p(t) = \Pr(x_j = 1)$ be the global fraction of variables currently in state 1.

Because each clause contains 2 variables, any given variable x_j participates in an average of 2α clauses. Specifically, it is connected to:

- α expected AND clauses
- α expected XOR clauses

When the algorithm selects a variable x_j to potentially flip, the outcome depends entirely on the current states of the *partner* variables in these local clauses. Let us classify these partner variables into four counts:

- A_1 : Number of AND clauses where the partner is 1.
- A_0 : Number of AND clauses where the partner is 0.
- X_1 : Number of XOR clauses where the partner is 1.
- X_0 : Number of XOR clauses where the partner is 0.

These local counts are not static averages; they are independent random variables. Assuming the network is well-mixed, the partner variables are drawn from the global distribution $p(t)$. Therefore, these counts follow Poisson distributions with means based on $p(t)$:

- $A_1 \sim \text{Poisson}(\alpha p)$
- $X_1 \sim \text{Poisson}(\alpha p)$
- $X_0 \sim \text{Poisson}(\alpha(1 - p))$

(Note: A_0 is irrelevant to the violation changes, as an AND clause with a 0 partner remains violated regardless of what x_j does)

The Single-Bit Flip Algorithm

Now we replace Juan's $(0, 0) \rightarrow (1, 1)$ pairwise flip with the algorithm's exact single-bit flip.

If $x_j = 0$ and we propose $0 \rightarrow 1$:

As we derived, the change in local violations is

$$\Delta V_{0 \rightarrow 1} = -A_1 + X_1 - X_0.$$

The algorithm strictly requires $\Delta V < 0$ to accept the flip.

$$\text{Accept } (0 \rightarrow 1) \leftrightarrow A_1 + X_0 > X_1$$

If $x_j = 1$ and we propose $1 \rightarrow 0$:

The change in local violations is

$$\Delta V_{1 \rightarrow 0} = A_1 + X_0 - X_1.$$

Again, the algorithm strictly requires $\Delta V < 0$.

$$\text{Accept } (1 \rightarrow 0) \leftrightarrow X_1 > A_1 + X_0$$

Probabilities

We now calculate the exact probabilities of these inequalities. Let us define a new random variable representing the combined effect to flip to 1:

$$Y = A_1 + X_0$$

Because the sum of two independent Poisson variables is also a Poisson variable, we add their means:

$$\mu_Y = \alpha p + \alpha(1 - p) = \alpha$$

Thus, the combined effect Y follows a $\text{Poisson}(\alpha)$ distribution, completely independent of the system state $p(t)$.

The effect for flipping to zero is simply X_1 , which follows $\text{Poisson}(\alpha p)$.

We can now write the exact, time-dependent acceptance probabilities for a single bit:

- $A_{0 \rightarrow 1}(t) = \Pr(Y > X_1)$
- $A_{1 \rightarrow 0}(t) = \Pr(X_1 > Y)$
- $A_{0 \rightarrow 1}(t) = \Pr(Z > 0)$
- $A_{1 \rightarrow 0}(t) = \Pr(Z < 0)$

Where $Z = Y - X_1$ is difference between two Poisson like Juan's result.

(Note: $\Pr(Z = 0)$ represents flips where $\Delta V = 0$, which your greedy algorithm strictly rejects).

Rate equation and steady state solution

Finally, we write the differential equation for the state $p(t)$.

If agents process new clauses at a rate r , and each clause contains 2 variables,

variables are tested for flips at a rate $2r$.

The rate of change of the fraction of ones is the flux of variables flipping $0 \rightarrow 1$ minus the flux of variables flipping $1 \rightarrow 0$:

$$\frac{dp}{dt} = 2r[(1 - p) \cdot \Pr(Z > 0) - p \cdot \Pr(Z < 0)]$$

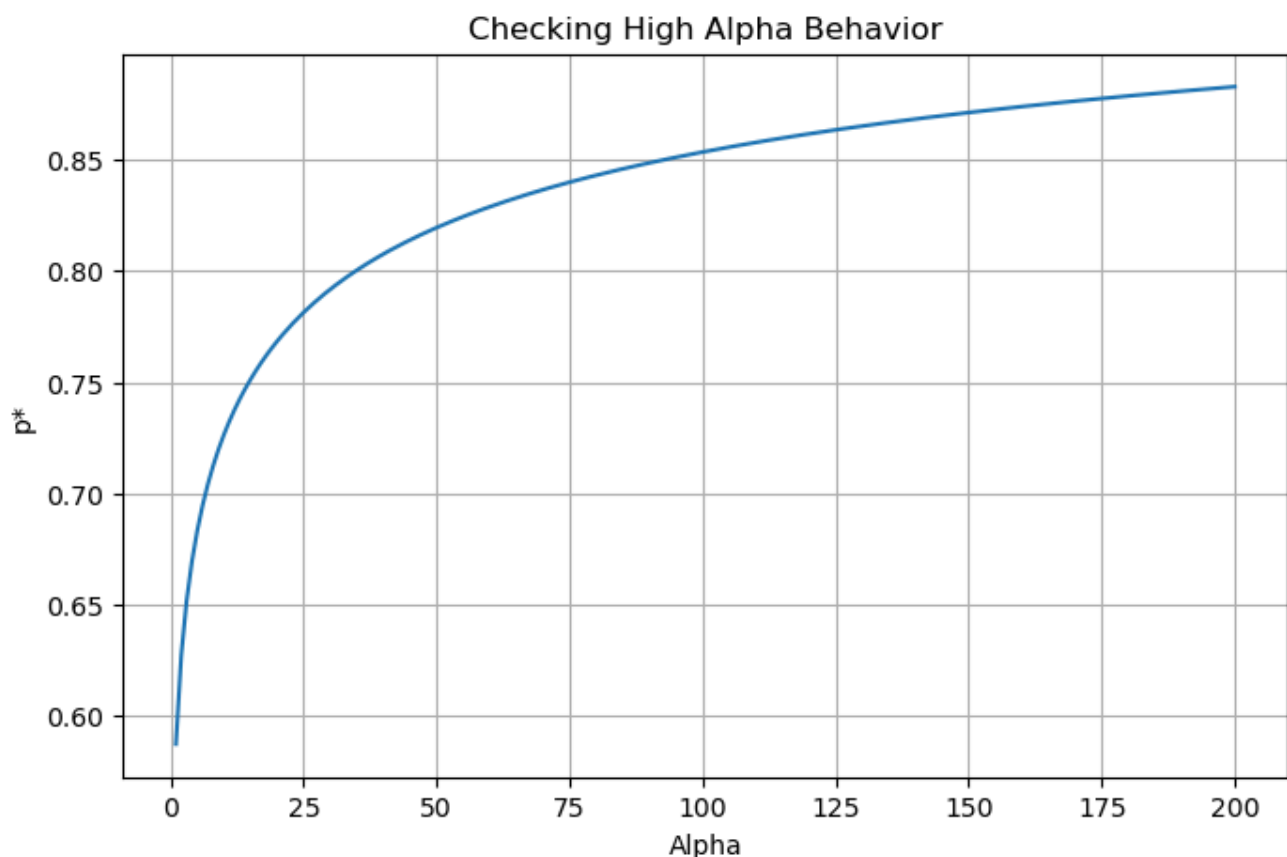
The Steady-State Condition

From the single-bit rate equation we just formulated, the system reaches steady state when $\frac{dp}{dt} = 0$:

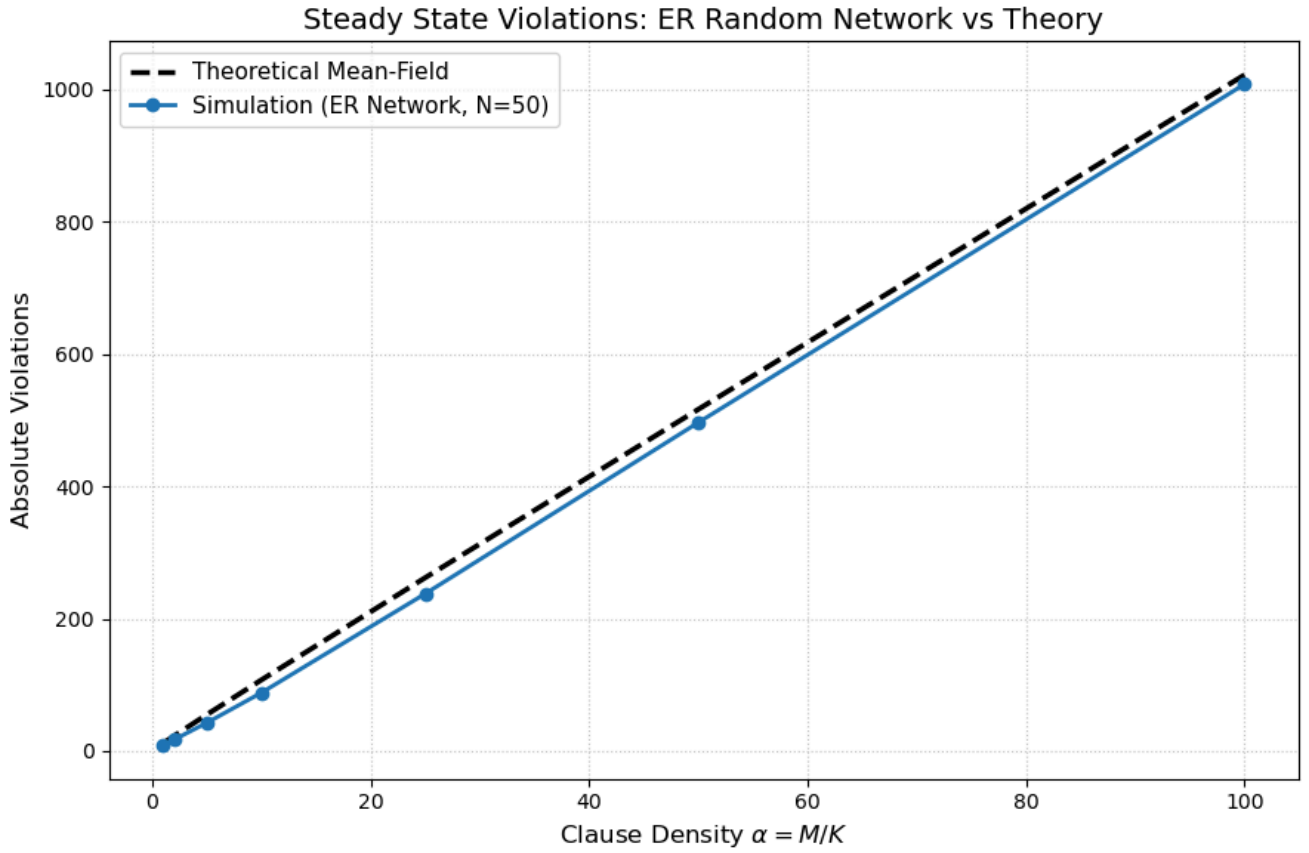
$$(1 - p^*) \Pr(Z > 0) = p^* \Pr(Z < 0)$$

Here, we use the Poisson variables $Y \sim \text{Poisson}(\alpha)$ and $X_1 \sim \text{Poisson}(\alpha p^*)$. We can solve for the equilibrium fraction of ones, p^* :

$$p^* = \frac{\Pr(Z > 0)}{\Pr(Z > 0) + \Pr(Z < 0)}$$



Comparing the violations obtained from p^* (above equation) and the simulation.



Dividing network nodes into classes of degree

Similar to k -core decomposition, we cannot easily solve N coupled ODEs, but we can solve a smaller system with $k = 1, \dots, k_{\max}$ equations.

We assume that all nodes with the exact same degree k are statistically equivalent. We replace the individual probability $p_i(t)$ with the class-average probability $p_k(t)$. Notice, this is like our first mean-field approximation. We know it is incomplete, but let us go on with it anyway.

In the model's implementation, the frequency at which an agent receives or observes clauses scales with their connectivity. A node processes observations at a baseline rate r_{obs} , and receives network interactions at a rate proportional to its degree k , scaled by an interaction parameter λ .

We define the total degree-dependent information gain rate R_k as:

$$R_k = 2 \left(r_{\text{obs}} + \lambda \frac{k}{\langle k \rangle} \right) \quad (1)$$

(We have a factor of 2, as a clause brings in two variables).

Substituting $p_k(t)$ and R_k into the microscopic equation yields the **generalized degree-based differential equation**:

$$\frac{dp_k}{dt} = R_k \left[(1 - p_k) A_k^{0 \rightarrow 1}(t) - p_k A_k^{1 \rightarrow 0}(t) \right] \quad (2)$$

We want to define these acceptance probabilities for the k -degree nodes. How do we do that?

These means are determined by the state of the agent's neighbors. If an agent's neighbors are highly accurate (holding correct bits), the agent is more likely to observe clauses that push them toward the correct state.

But how do we know the state of an agent's neighbors? In an uncorrelated (well-mixed) network, we can assume that any neighbor an agent observes is simply a random node drawn from the entire population.

Therefore, the expected number of each clause type an agent observes depends **only on the global average** $p(t)$, not on the agent's specific degree k . Because the means of the Poisson variables depend exclusively on the global state $p(t)$, the acceptance probabilities lose all local network dependence. For every agent in the network, regardless of their position or degree, the evaluation of the inequality yields the exact same statistical result.

The acceptance probabilities become universal, time-dependent functions driven entirely by $p(t)$:

$$A_k^{0 \rightarrow 1}(t) = \Pr(Z > 0) \quad (3a)$$

$$A_k^{1 \rightarrow 0}(t) = \Pr(Z < 0) \quad (3b)$$

Thus, we can substitute these values back into our degree-based equation:

$$\frac{dp_k}{dt} = R_k \left[(1 - p_k) \Pr(Z > 0) - p_k \Pr(Z < 0) \right] \quad (4)$$

We can rewrite this in a more simplified form:

$$\frac{dp_k}{dt} = R_k \left[\Pr(Z > 0) - p_k(t) (\Pr(Z > 0) + \Pr(Z < 0)) \right] \quad (5)$$

The term in the square brackets is time-dependent (driven by the global $p(t)$), whereas R_k strictly provides the network dependence.

The initial derivative for class k at $t = 0$ is:

$$\left. \frac{dp_k}{dt} \right|_{t=0} = R_k F_0$$

where $F_0 = \Pr(Z_0 > 0) - p_0 (\Pr(Z_0 > 0) + \Pr(Z_0 < 0))$.

We construct a first-order Taylor expansion for the early-time trajectory:

$$p_k(t) \approx p_0 + \left. \frac{dp_k}{dt} \right|_{t=0} t$$

The difference in knowledge between a hub of degree k and a periphery node of degree k' over early time t is the difference between their Taylor expansions:

$$\begin{aligned} \Delta_{k,k'}(t) &\approx \left. \frac{dp_k}{dt} \right|_{t=0} t - \left. \frac{dp_{k'}}{dt} \right|_{t=0} t \\ \Delta_{k,k'}(t) &\approx 2F_0 \lambda \left(\frac{k - k'}{\langle k \rangle} \right) t \end{aligned} \tag{6}$$

Therefore, the early-time learning trajectories diverge linearly, driven strictly by the difference in degree $(k - k')$. We can numerically verify this linearity condition. Notice, this is still strictly under the uniform global distribution assumption, but we do see linear dynamics in the immediate time for various degrees.